



DEALER BEST PRACTICE SERIES

3500 Crankshaft Installing Guide Block

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Many Trakindo branches overhaul engines and some technicians have found it difficult to install the crankshaft. This can affect the efficiency of the installation, expose the component to potential damage, and reduce cleanliness and safety. To improve the crankshaft installation, Trakindo has created a standard operating procedure to install the crankshaft using guide blocks.

2.0 Best Practice Description

Many times when installing a crankshaft to a cylinder block, it will collide with the cylinder block. This Best Practice helps the technician avoid the potential collision when using the proper lifting equipment for this installation. The guide blocks presented in this document assist in guiding crankshaft's main journals into the blocks main bearing.

Tools			
Tool	Part Number	Part Description	Qty
A	FT	Crankshaft Guide Block	4
B	2N-5014	Bolt	4

Note : Drawing FT Crankshaft guide block can be view in TSD-HO Database

Figure 1: Parts needed for guide blocks.



Figure 2: Guide blocks and bolts.

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Figure 3: Guide block location on cylinder block.



Figure 4: Guide block in front and rear cylinder block bearing webs.

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3.0 Implementation Steps

- Fabricate the Guide Block from solid rubber or polyurethane.
- Integrate use of the Guide Blocks into the standard operating procedure.
 - o Install the guide blocks after the upper half of the main bearings are installed.
 - o Assure shaft is handled with crane and appropriate strap/chain/fixture.
- Train employees to use the new procedures/equipment.
- Review process/equipment performance once established.

4.0 Benefits

Safety and Installation Time

- Well designed/organized installation will improve the time to install and safety.

Quality

- Guide blocks allow crankshaft installation with little or no journal denting opportunities.
- A more effective/simplified installation process will promote higher technician professionalism, higher standards, and increased brand image.

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5.0 Resources Required

Tools

- 4 x (89 x 57.2 x 80) mm fabricated solid rubber/polyurethane
- 4 x 2N-5014 Bolt

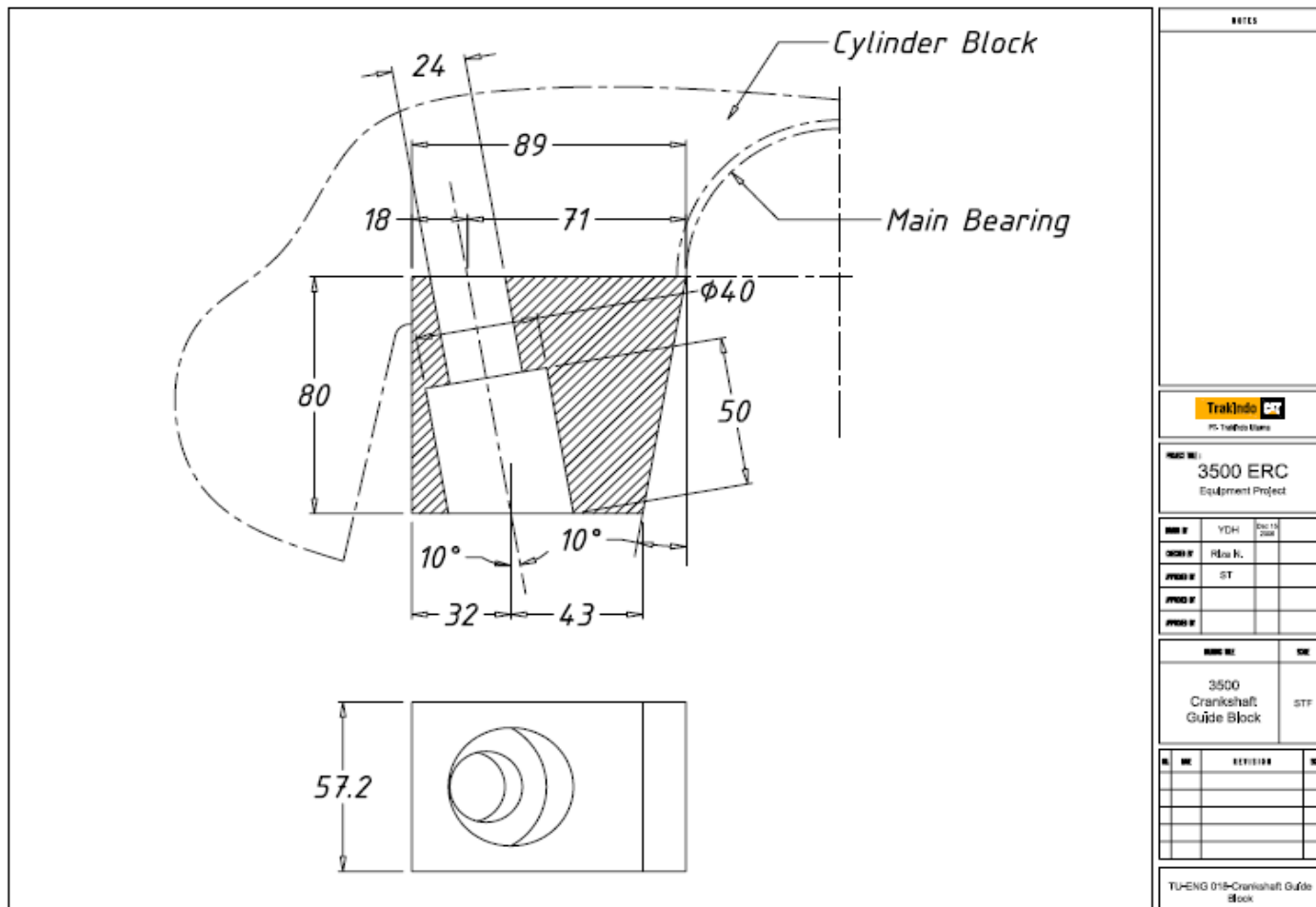


Figure 5: Drawing of guide block to be fabricated.

People

- Establish the new process, layout, and equipment improvement through the team.
- Remove replaced items from less effective crankshaft installation processes. Sustain the gains.

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6.0 Supporting Attachments / References

Working Instructions – Install Crankshaft Using Guide Block
(WIN Install Crankshaft Using Guide Block.doc)

Cat tooling for C175 Crank Installation Guides (4) 417-2349
Photos of use at LEC attached

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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